AWS CLOUD COMPUTING PROJECT

Highly Available Web Application Using Amazon Web Services

Project Documentation | Days 51–57

Submitted by: _____

College / University: _____

Department: _____

Academic Year: _____

AWS Region: us-east-1

Abstract

This project demonstrates the deployment and operation of a highly available web application on Amazon Web Services (AWS). The solution uses an Amazon VPC, public and private subnets across two Availability Zones, an Internet Gateway, Amazon EC2, an Application Load Balancer, an Auto Scaling Group, Amazon CloudWatch, and Amazon S3. Two EC2 instances serve the web application. The ALB distributes HTTP requests to registered targets, while Auto Scaling maintains the configured instance capacity. CloudWatch displays CPU metrics and evaluates an alarm. Amazon S3 stores a website backup object.

1. Project Objectives

- Create an isolated AWS network using a VPC and subnets.
- Configure internet connectivity through an Internet Gateway and route tables.
- Deploy a web application on EC2 using a Launch Template.
- Distribute HTTP traffic through an Application Load Balancer.
- Maintain two EC2 instances using an Auto Scaling Group.
- Monitor CPU utilization with CloudWatch.
- Store a website backup object in Amazon S3.

2. AWS Services and Roles

Service / component	Purpose
VPC	Isolated project network.
Subnets	Public/private network segments across two AZs.
Internet Gateway	Internet connectivity for public subnets.
Route Tables	Control network routing.
Security Groups	Control inbound traffic.
EC2	Runs Apache and the sample website.
Launch Template	Defines EC2 launch configuration.
ALB / Target Group	Routes HTTP traffic to healthy EC2 targets.
Auto Scaling Group	Maintains desired instance count.
CloudWatch	Displays CPU metrics and evaluates alarms.
S3	Stores website backup object.

3. Network Design and Configuration

VPC: HA-WebApp-VPC **CIDR:** 10.0.0.0/16 **Region:** us-east-1

Subnet	Type	IPv4 CIDR
Public-Subnet-A	Public	10.0.1.0/24
Private-Subnet-A	Private	10.0.2.0/24
Public-Subnet-B	Public	10.0.3.0/24
Private Subnet B	Private	10.0.4.0/24

The HA-WebApp-IGW is attached to the VPC. The public route table contains 0.0.0.0/0 pointing to the Internet Gateway. The private route table contains the local VPC route only. No NAT Gateway was deployed to reduce lab costs. In this training design, the ALB and EC2 instances were placed in public subnets; the EC2 security group is intended to accept

web traffic only from the ALB security group. A production design would normally place application instances in private subnets.

4. Application Deployment

The Launch Template HA-WebApp-Template uses Amazon Linux and a t3.micro instance type. EC2 User Data installs and starts Apache HTTP Server and creates the sample index.html page. The HA-WebApp-ASG uses the template to maintain two EC2 instances. The internet-facing ALB HA-WebApp-ALB uses an HTTP listener on port 80 and forwards traffic to HA-WebApp-TG.

5. Auto Scaling and High Availability

The supplied Auto Scaling Group screenshot shows desired capacity 2, minimum 2, maximum 2, and status At desired capacity. The two EC2 instances are in separate Availability Zones (us-east-1a and us-east-1b). The recovery test was reported successful: Auto Scaling restored the group to two running instances, both targets became healthy, and the website continued to work through the ALB.

6. Monitoring with CloudWatch

CloudWatch was used to view EC2 CPUUtilization and create the alarm HA-WebApp-High-CPU. The supplied alarm screenshot shows the alarm in OK state and a threshold line at 80 percent. The CPU graph and alarm state can be reviewed to identify elevated utilization.

7. Amazon S3 Backup

The bucket ha-webapp-backup-vinay-2026 contains the uploaded index.html object. S3 Versioning was part of the practical backup workflow as reported during the session. The supplied screenshot shows the object list but does not display version history, so it is not presented as screenshot proof of multiple versions.

8. Security and Cost Considerations

- ALB security group allows HTTP traffic on port 80.
- EC2 security group restricts HTTP ingress to the ALB security group.
- Keep S3 Block Public Access enabled unless a reviewed use case requires otherwise.
- Use IAM roles and least-privilege permissions.
- EC2, EBS, ALB, public IPv4 addresses, S3, and CloudWatch features may incur charges. Review Billing and credits regularly.
- Stopping EC2 does not remove attached EBS storage charges. Delete unused resources when the lab is finished.

9. Testing and Results

- VPC and four subnets verified as available.
- Internet Gateway attachment and public default route verified.
- Two EC2 instances running with 3/3 status checks passed.
- Auto Scaling Group at desired capacity of two.
- ALB active and internet-facing.
- Target group showed two healthy targets and zero unhealthy targets.
- Website opened successfully through the ALB DNS name.
- Recovery test restored two running instances and healthy targets.
- CloudWatch CPU alarm created and shown in OK state.

- index.html uploaded to the S3 bucket.

10. Conclusion

The project demonstrates a practical AWS deployment workflow combining network design, EC2 provisioning, load balancing, Auto Scaling, monitoring, and object storage. The successful website test and healthy target group show that the application was reachable through the load balancer at the time of testing.

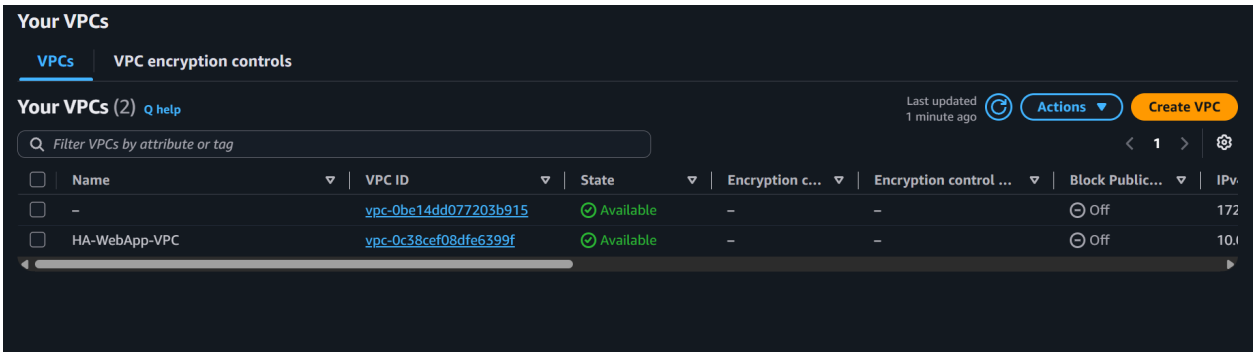
11. Future Enhancements

- Move application EC2 instances into private subnets and configure controlled outbound access.
- Configure HTTPS using AWS Certificate Manager.
- Automate deployment using GitHub Actions.
- Automate S3 backups and test recovery.
- Configure CloudWatch alarm notifications.
- Manage infrastructure using Terraform or CloudFormation.

12. Evidence Screenshots

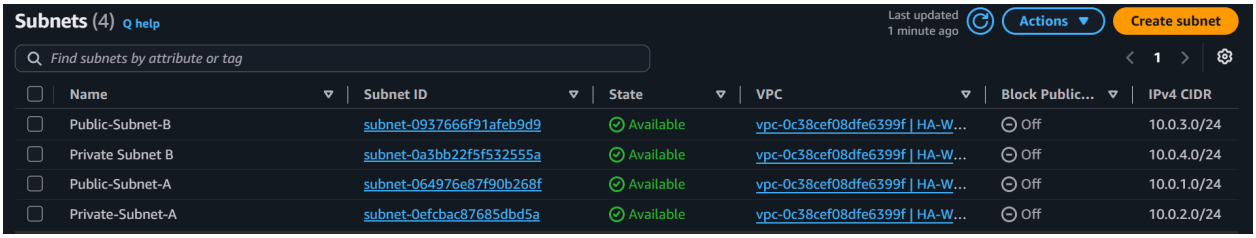
Screenshots below were supplied during the practical work. Each caption describes the visible console state.

Figure 1. VPC configuration



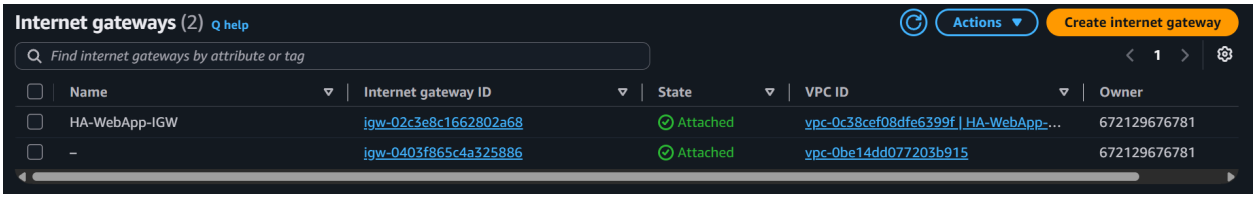
The HA-WebApp-VPC is present and available in the AWS account.

Figure 2. Subnet configuration



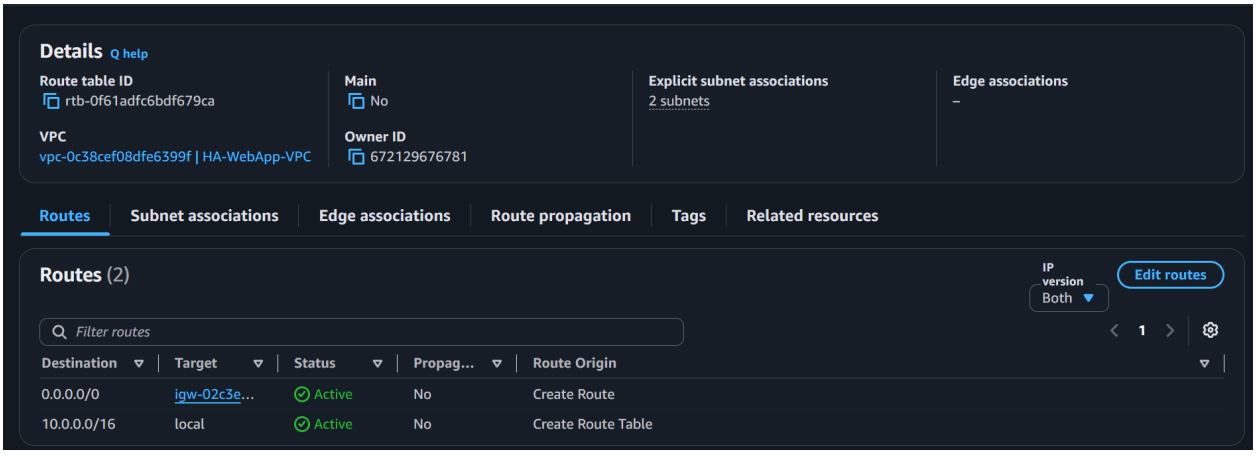
Four subnets are configured across two Availability Zones: two public and two private.

Figure 3. Internet Gateway



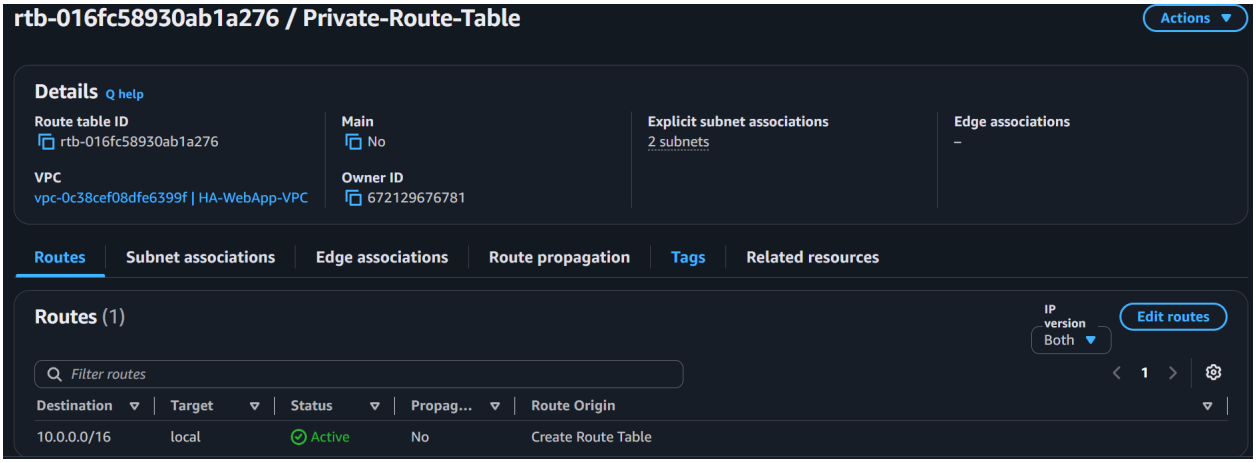
The HA-WebApp-IGW is attached to the project VPC.

Figure 4. Public route table



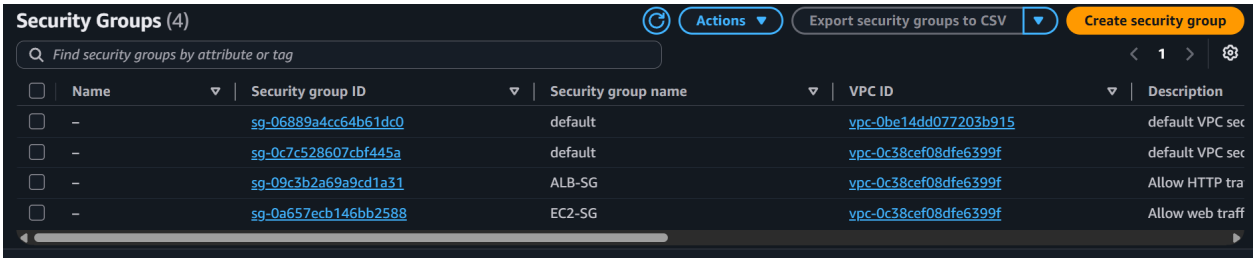
The public route table includes the default route 0.0.0.0/0 to the Internet Gateway.

Figure 5. Private route table



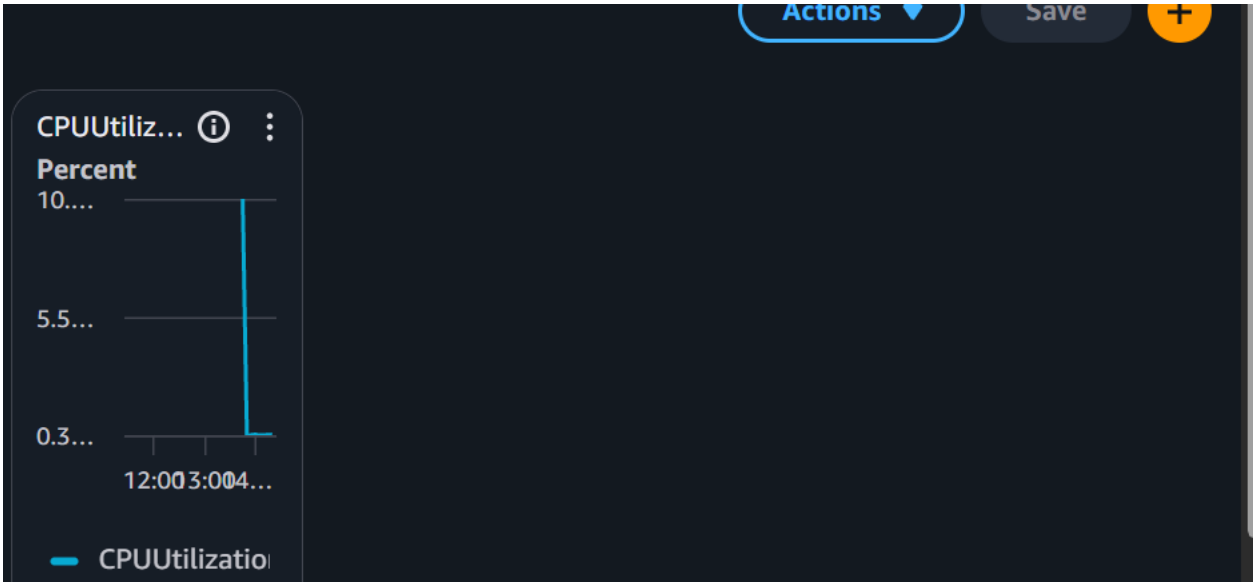
The private route table contains the VPC local route; no NAT Gateway route is configured.

Figure 6. Security groups



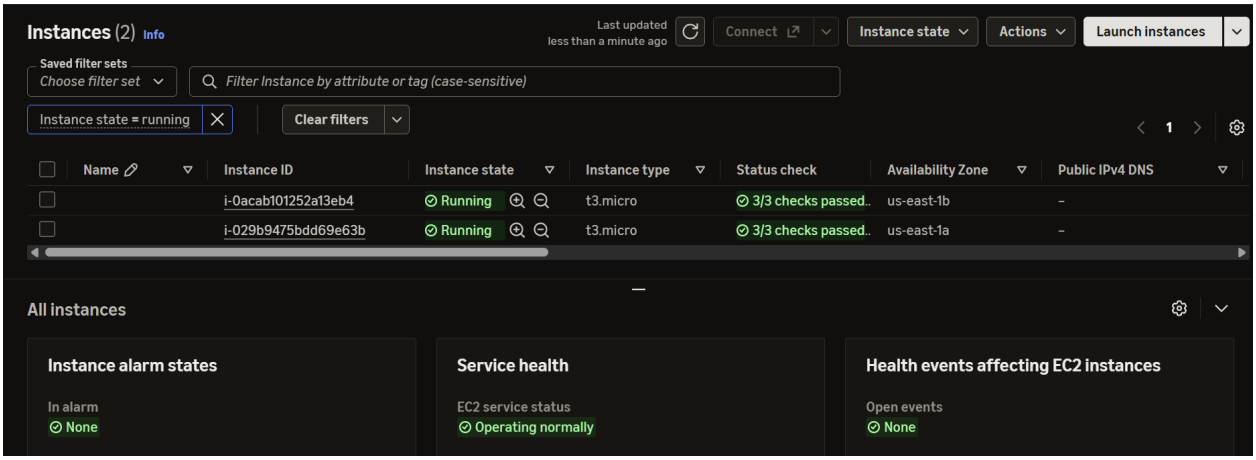
The ALB-SG and EC2-SG security groups are configured in the project VPC.

Figure 7. CloudWatch CPU metric



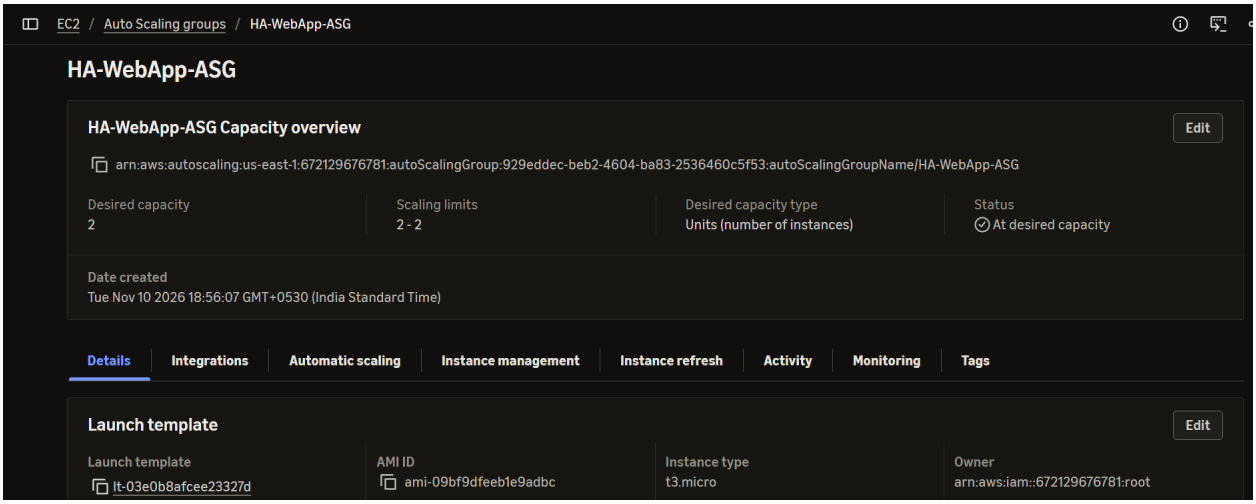
A CPUUtilization metric graph is displayed in CloudWatch.

Figure 8. EC2 instances



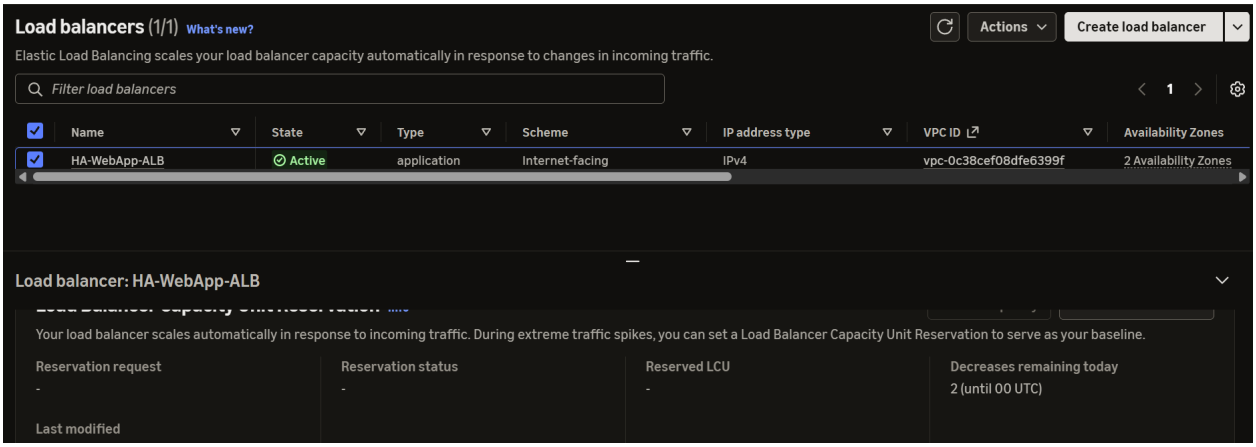
Two t3.micro EC2 instances are running and both show 3/3 status checks passed.

Figure 9. Auto Scaling Group



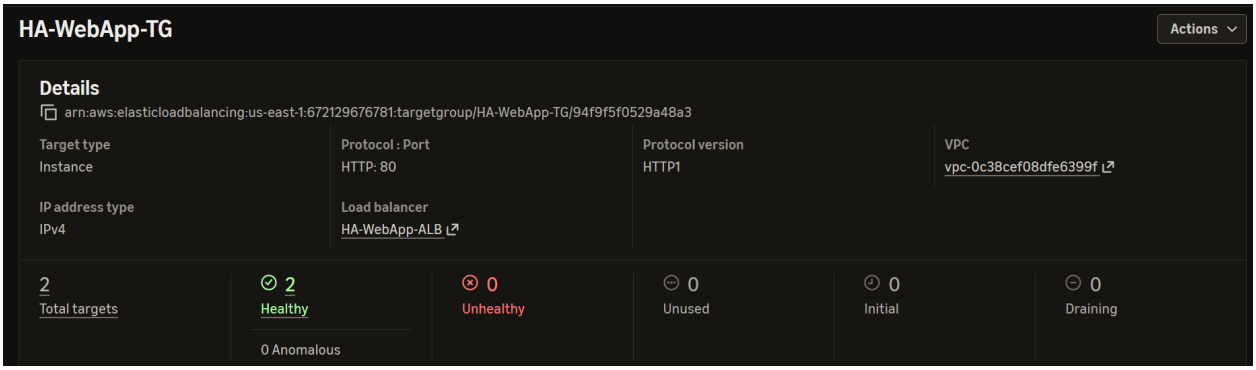
HA-WebApp-ASG shows desired capacity 2, minimum 2, maximum 2, and At desired capacity.

Figure 10. Application Load Balancer



HA-WebApp-ALB is active, internet-facing, and configured across two Availability Zones.

Figure 11. Target Group health



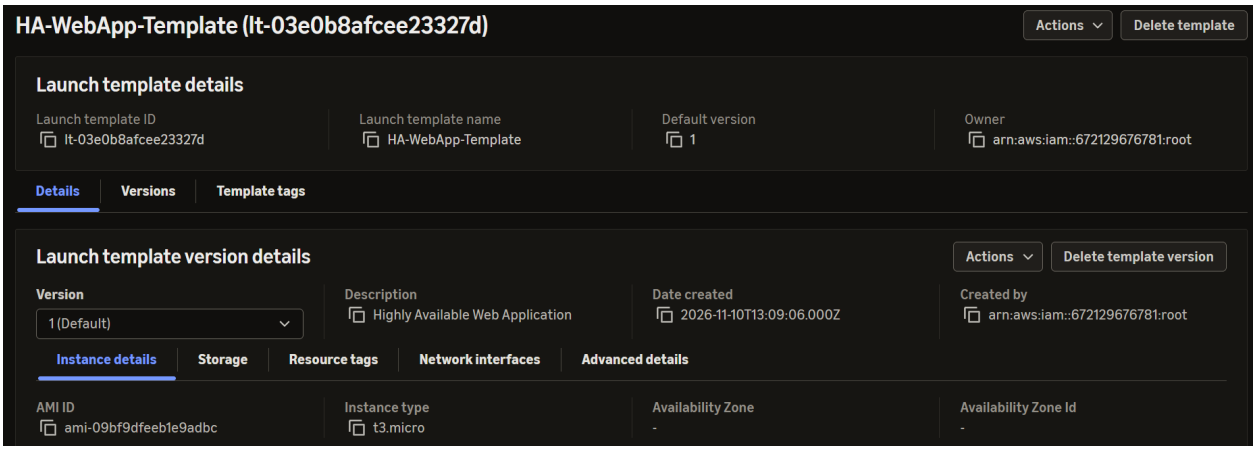
HA-WebApp-TG shows two healthy targets and zero unhealthy targets.

Figure 12. Deployed website



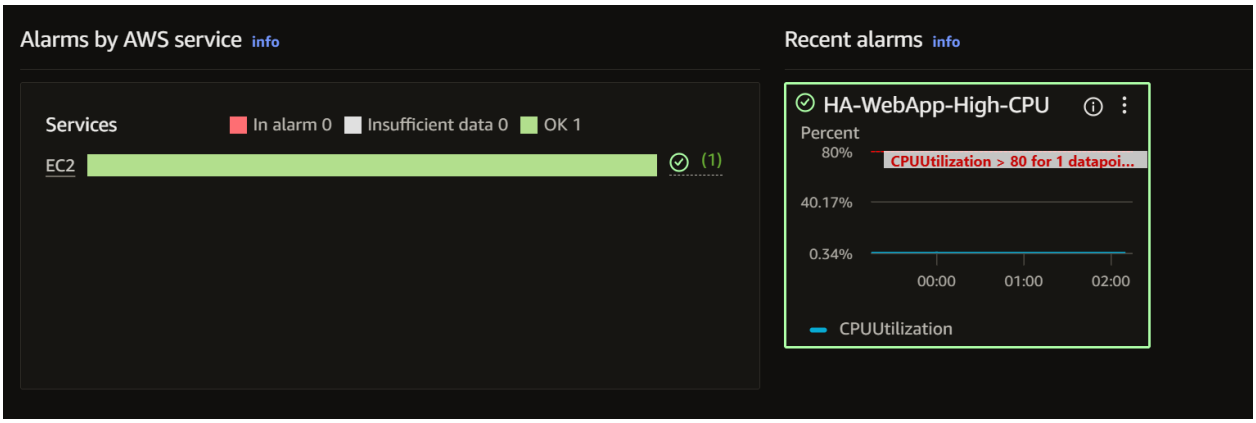
The sample highly available web application page loads successfully.

Figure 13. Launch Template



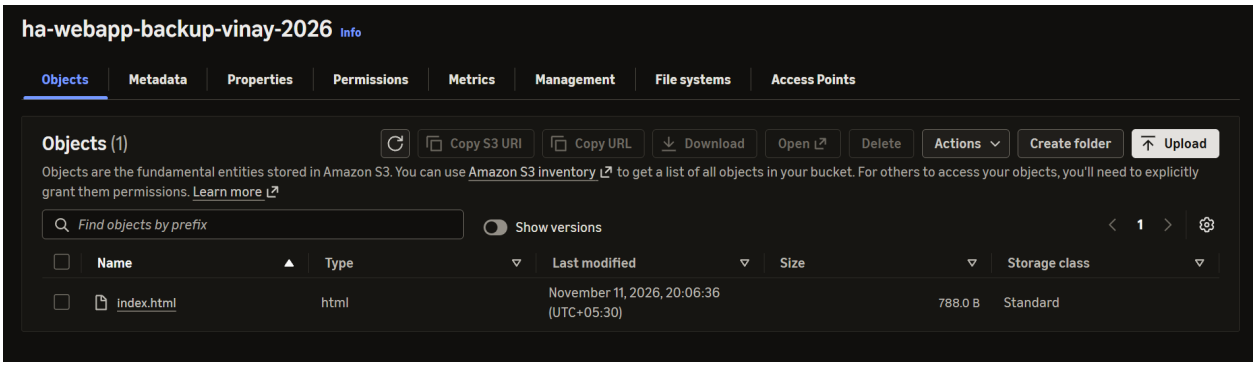
HA-WebApp-Template is configured with a t3.micro instance type.

Figure 14. CloudWatch alarm



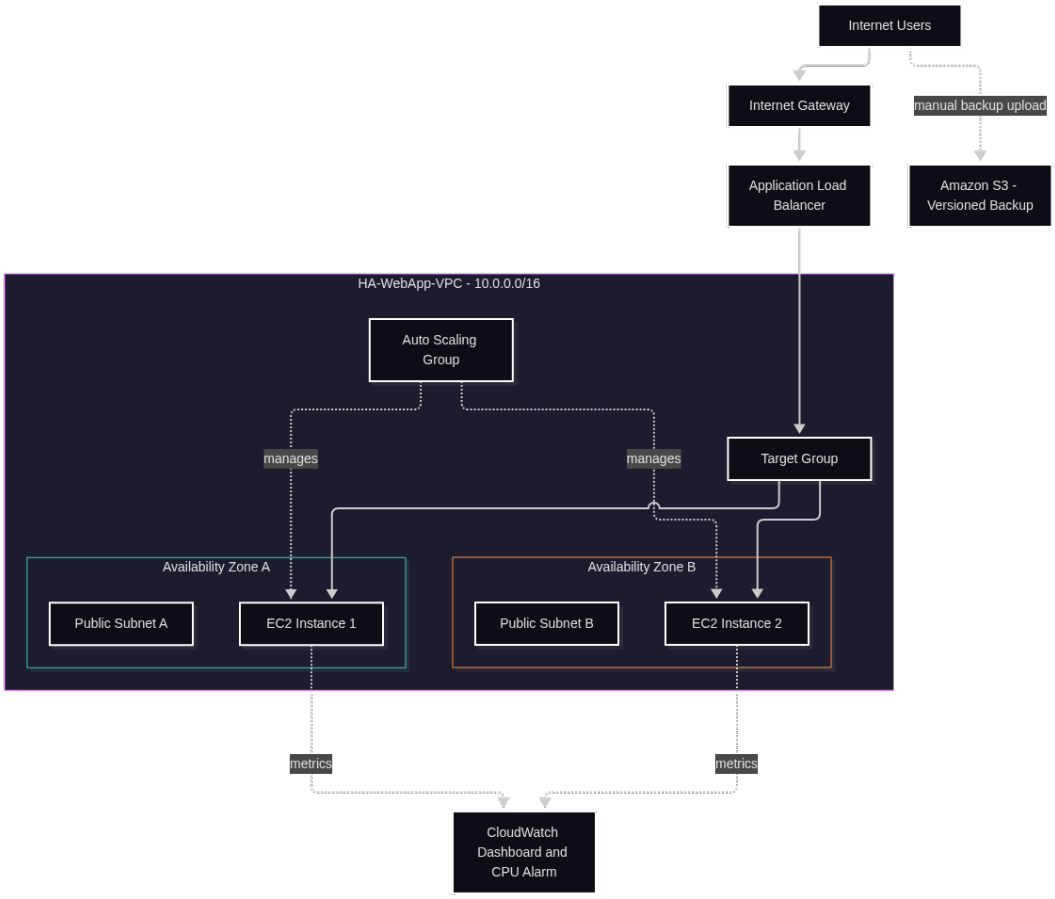
HA-WebApp-High-CPU is shown in OK state. The alarm graph displays the CPU threshold.

Figure 15. S3 backup object



The S3 bucket contains the uploaded index.html backup object.

Figure 16. Architecture diagram



The final architecture diagram summarizes traffic flow, EC2 instances, Auto Scaling, CloudWatch, and S3 backup.